EMS611U: Conceptual Design of Aircraft

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Week 4

Aerofoil & Geometry Selection

Aerofoil Design

What should we consider selecting an aerofoil?

In the past, the designer would select an aerofoil from such a catalog by considering:

- Aerofoil drag during cruise
- Stall and pitching-moment characteristics,
- The thickness available for structure,
- Fuel
- The ease of manufacture.

With today's computational aerofoil design capabilities, it is becoming common for the aerofoil shapes for a wing to be custom-designed.

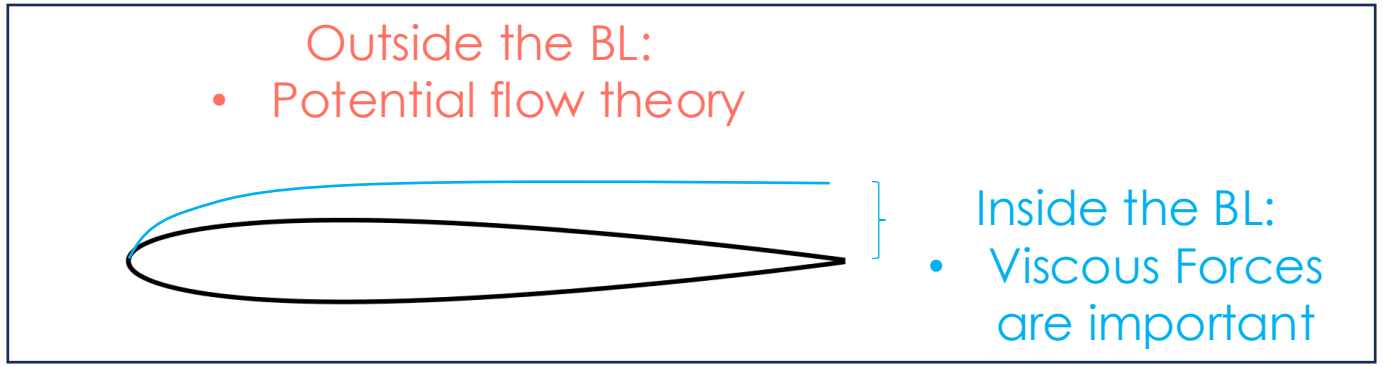
Methods have been developed for designing an aerofoil such that :

- The pressure differential between the top and bottom of the aerofoil quickly reaches a maximum value attainable without flow separation.
- Toward the rear of the aerofoil, various pressure recovery schemes are employed to prevent separation near the trailing edge.

Larger lift or **Reduced wetted area** for a required amount of lift → Lower drag for a required lift → Overall: Larger L/D

Aerofoil Design

Most aerofoils are designed using numerical codes based around potential flow theory with boundary layer corrections, but one can also use wind tunnel data or inverse design methods.

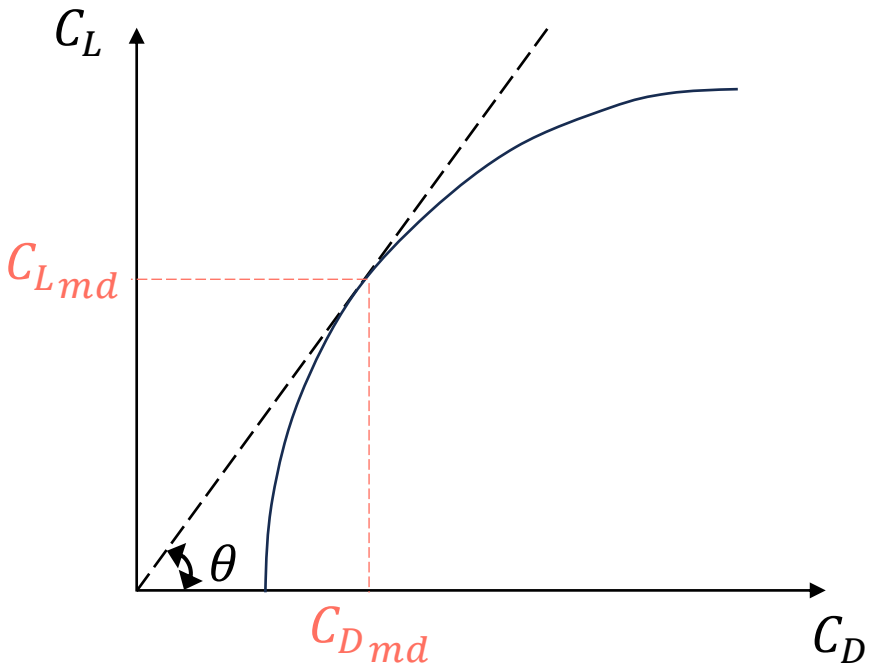


Most aerofoils are designed for a specific design point, such as:

- Maximum Lift
- Maximum Thickness
- Transonic Flight
- Laminar Boundary Layer
- Low Reynolds Number
- Low Pitching Moment

Design Lift Coefficient

- The first consideration in initial aerofoil selection is the "design lift coefficient."
- This is the lift coefficient at which the aerofoil has the best (L/D).



$$\tan(\theta) = \frac{C_{L_{md}}}{C_{D_{md}}} = \left(\frac{L}{D}\right)_{max}$$

In subsonic flight a well-designed aerofoil operating at its design lift coefficient has a drag coefficient that is little more than skin-friction drag.

The aircraft should be designed so that it flies the design mission at or near the design lift coefficient to maximize the aerodynamic efficiency.

Design Lift Coefficient

$W = L = q\,S\,C_L \cong qSC_l$ (First Approximation)

- C_L: Wing Lift Coefficient
- C_l: Aerofoil Lift Coefficient

$$C_l = \frac{1}{q}\left(\frac{W}{S}\right)$$

- $\frac{W}{S}$: Wing Loading

$$q = f(V, h)$$

C_l can be calculated for the velocity and altitude of the design.

During the mission fuel is burned: $\frac{W}{S} \downarrow$ $q \downarrow$ $\rightarrow h \uparrow$

"cruise climb flight"

↓

"maximum range"

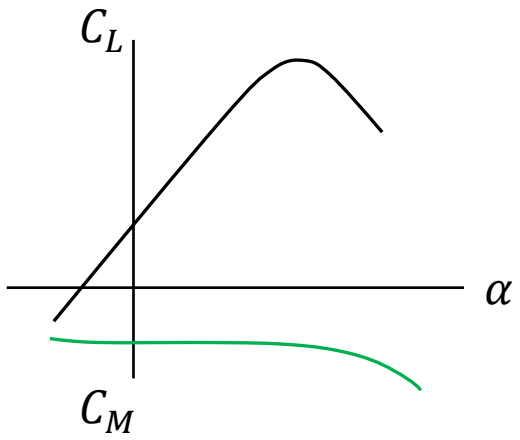
- In actual practice, a design lift coefficient usually will be based upon past experience, and for most types of aircraft will be around "0.5".
- In fact, the initial selection of the aerofoil is often simply based upon prior experience or copied from some successful design.

Stall

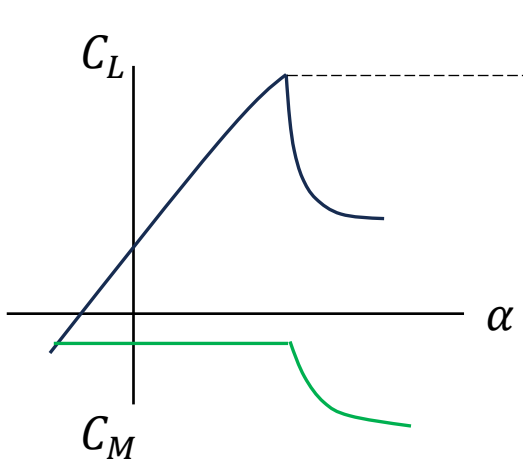
- Stall characteristics play an important role in aerofoil selection.
- Some aerofoils exhibit a gradual reduction in lift during a stall, whereas others show a violent loss of lift, accompanied by a rapid change in pitching moment. This difference reflects the existence of three entirely different types of aerofoil stall.

Stall Patterns:

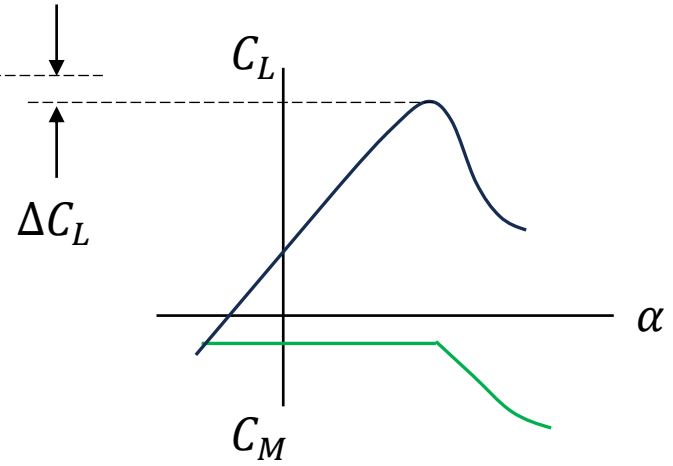
Gradual Stall



Abrupt Stall



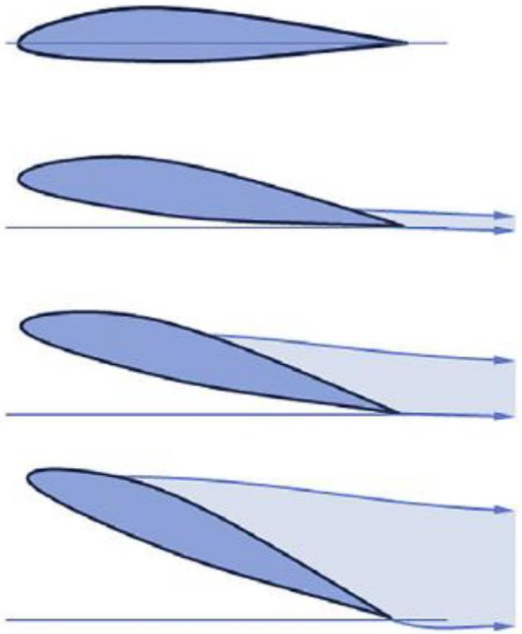
Thin Aerofoil

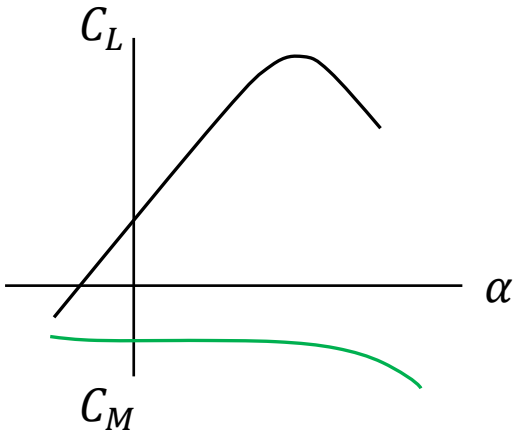


Stall

Gradual Stall

$\alpha \uparrow$

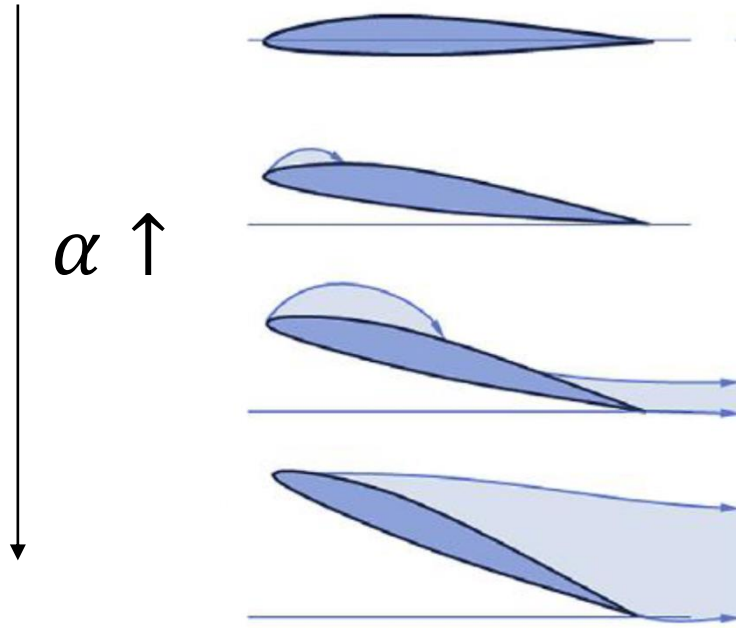


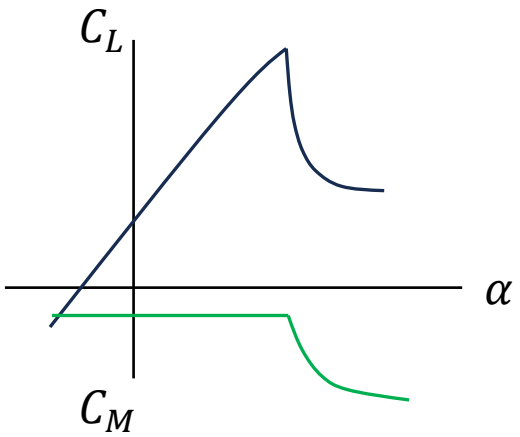


- This is a desirable pattern and occurs when separation is initiated at the trailing edge of the aerofoil; the remainder maintains the pressure differential.
- At around 10° the boundary layer begins to separate, starting at the trailing edge and moving forward as the angle of attack is further increased.
- The aircraft approaches stall gradually, giving the pilot enough time to take corrective action. The forgiving and gentle nature of this stall is ideal for an trainee pilot.
- The loss of lift is gradual. The pitching moment changes only a small amount.
- The type of aerofoil that experiences this type of stall has a generously rounded LE, providing smooth flow negotiation and t/c greater than about %14. However, it does not necessarily offer other desirable performance characteristics.

Stall

Abrupt Stall

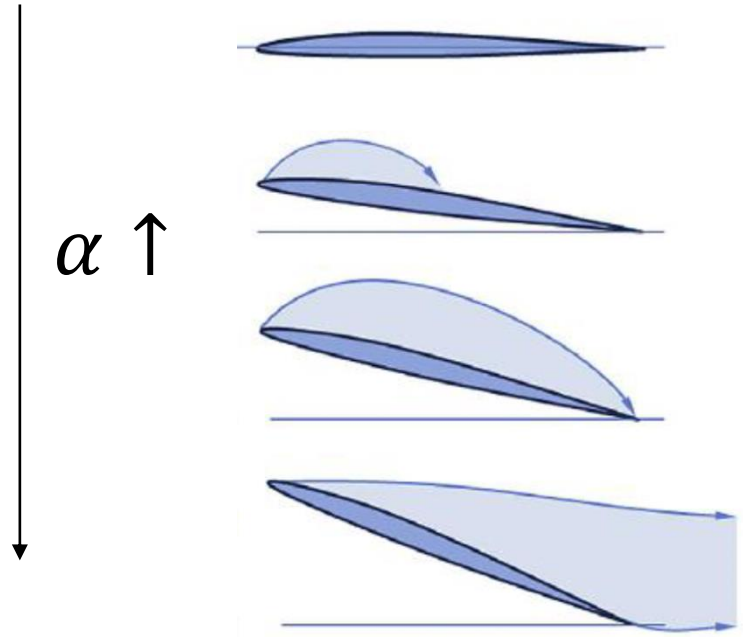


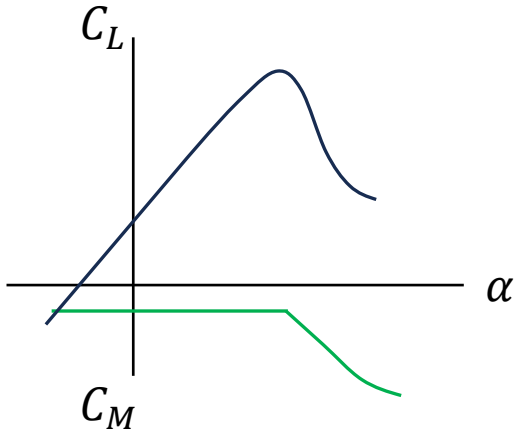


- Thinner aerofoils stall from the leading edge. If the aerofoil is of moderate thickness (about 6–14%).
- The flow separates near the nose at a very small angle of attack but immediately reattaches itself so that little effect is felt. At some higher angle of attack the flow fails to reattach, which almost immediately stalls the entire aerofoil.
- This causes an abrupt change in lift and pitching moment.
- *Aerofoils with a sharper LE, such as those found on higher-performance aircraft, tend to exhibit this type of behavior.*

Stall

Thin Aerofoil





- Very thin aerofoils exhibit another form of stall, t/c < %6.
- As before, the flow separates from the nose at a small angle of attack and reattaches almost Immediately.
- For a very thin aerofoil this "bubble" continues to stretch towards the trailing edge as the angle of attack is increased.
- At the angle of attack where the bubble stretches all the way to the trailing edge, the aerofoil reaches its maximum lift.
- Beyond that angle of attack, the flow is separated over the whole aerofoil, so the stall occurs.
- The loss of lift is smooth, but large changes in pitching moment are experienced.

Stall

The main aerofoil design features which affect section stall, therefore the maximum lift coefficient are:

1.Thickness ratio
2.Shape of the mean line (camber)
3.Leading edge shape and leading edge radius.

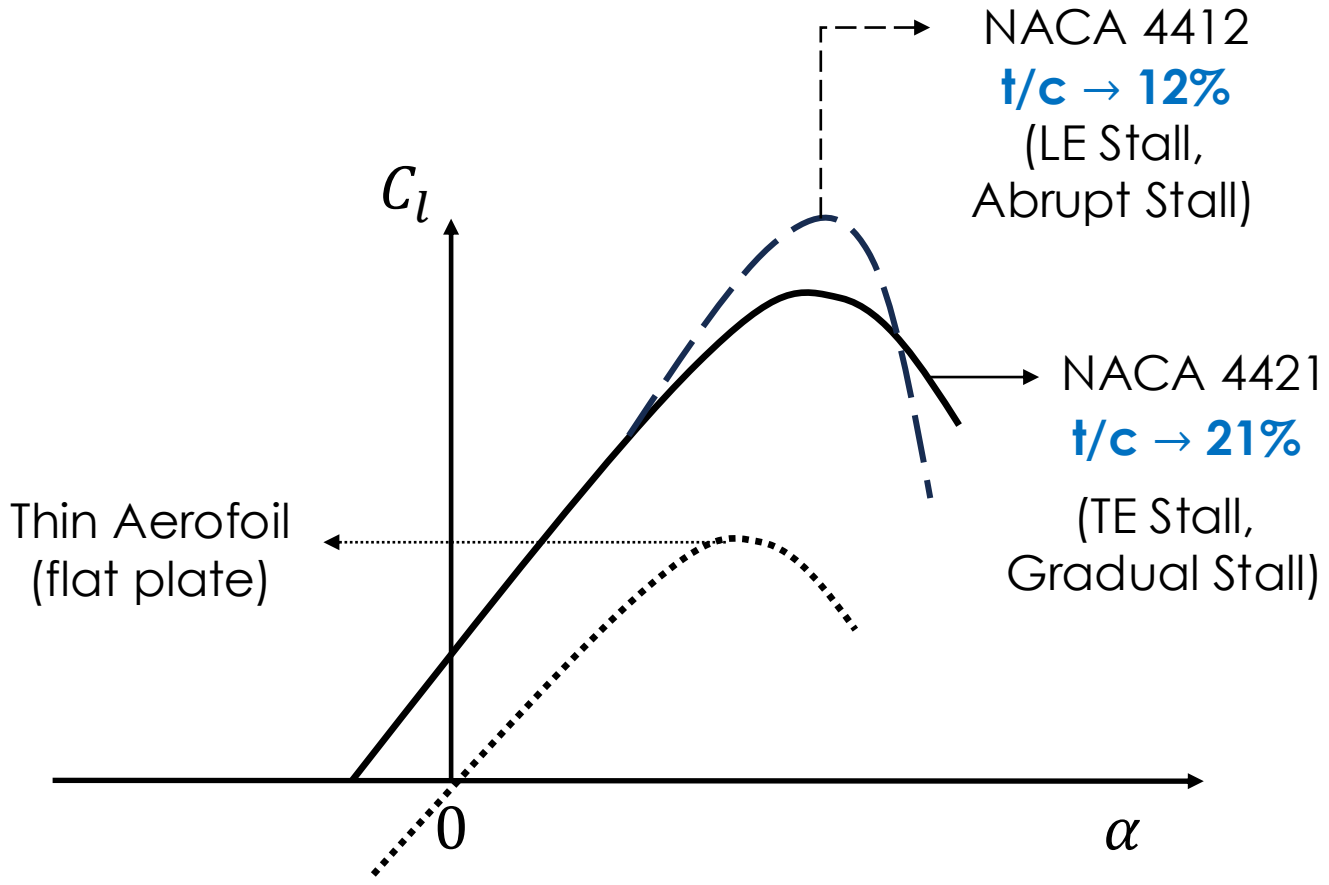
Maximum Lift Characteristics

Aerofoil Thickness Ratio

Aerofoil thickness ratio has a direct effect on:

- Maximum lift
- Stall characteristics
- Drag
- Structural weight.

Both NACA 4412 and NACA 4421 have same shape of mean camber line.



Notes:

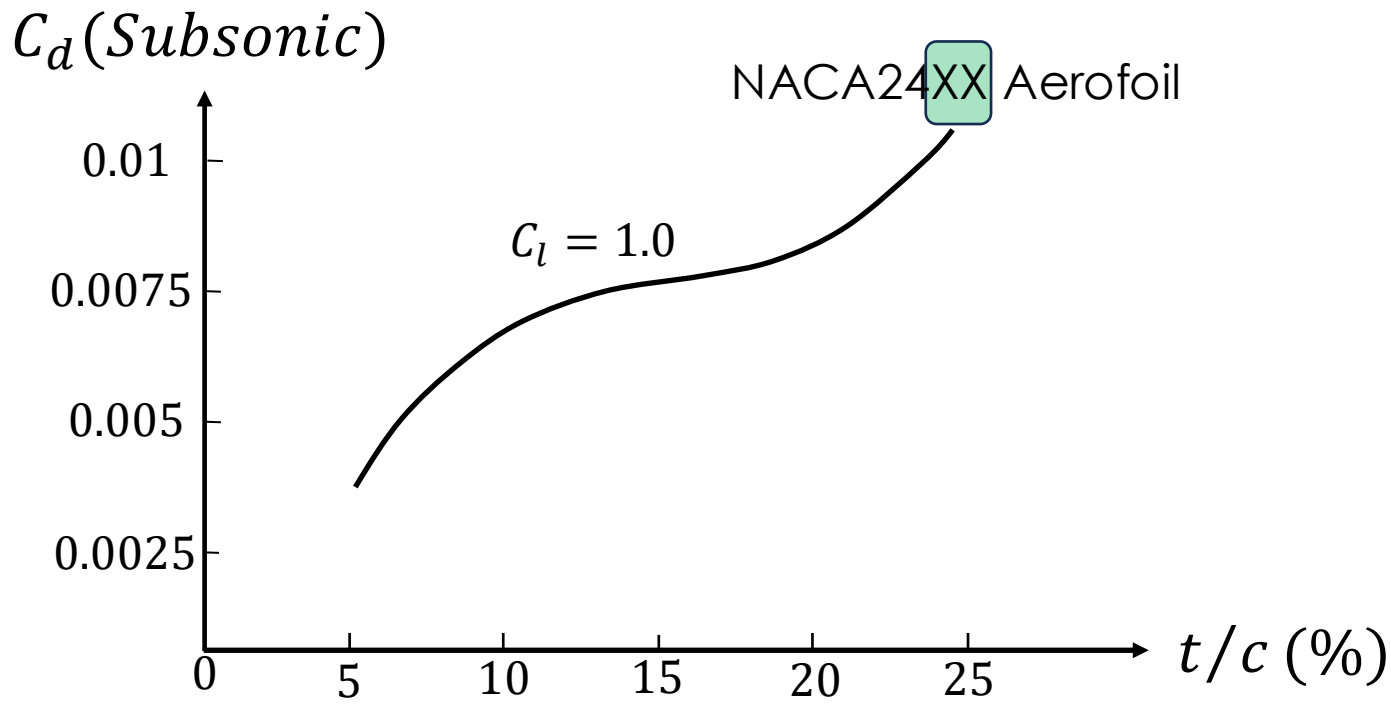
- **Leading edge stall** shows rapid drop of lift curve near maximum lift.
- **Trailing edge stall** shows gradual bending-over of lift curve at maximum lift, "soft stall"
- High $C_{l_{max}}$ for aerofoils with leading edge stall
- Flat plate stall exhibits poorest behavior, early stalling

Maximum Lift Characteristics

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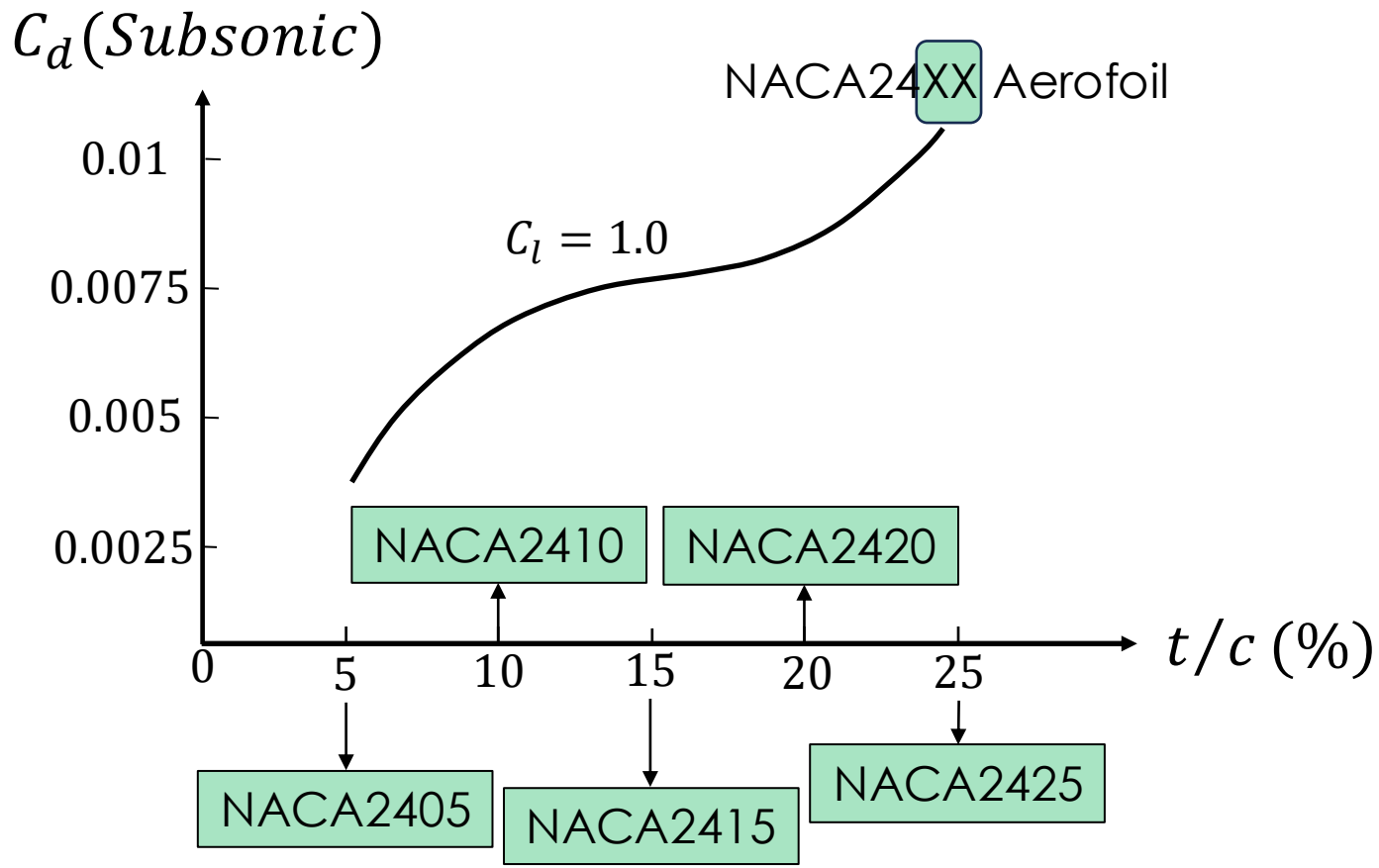
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- Structural weight.

Notes:

- As the thickness ratio increases, drag increases, due to increased separation.



Maximum Lift Characteristics

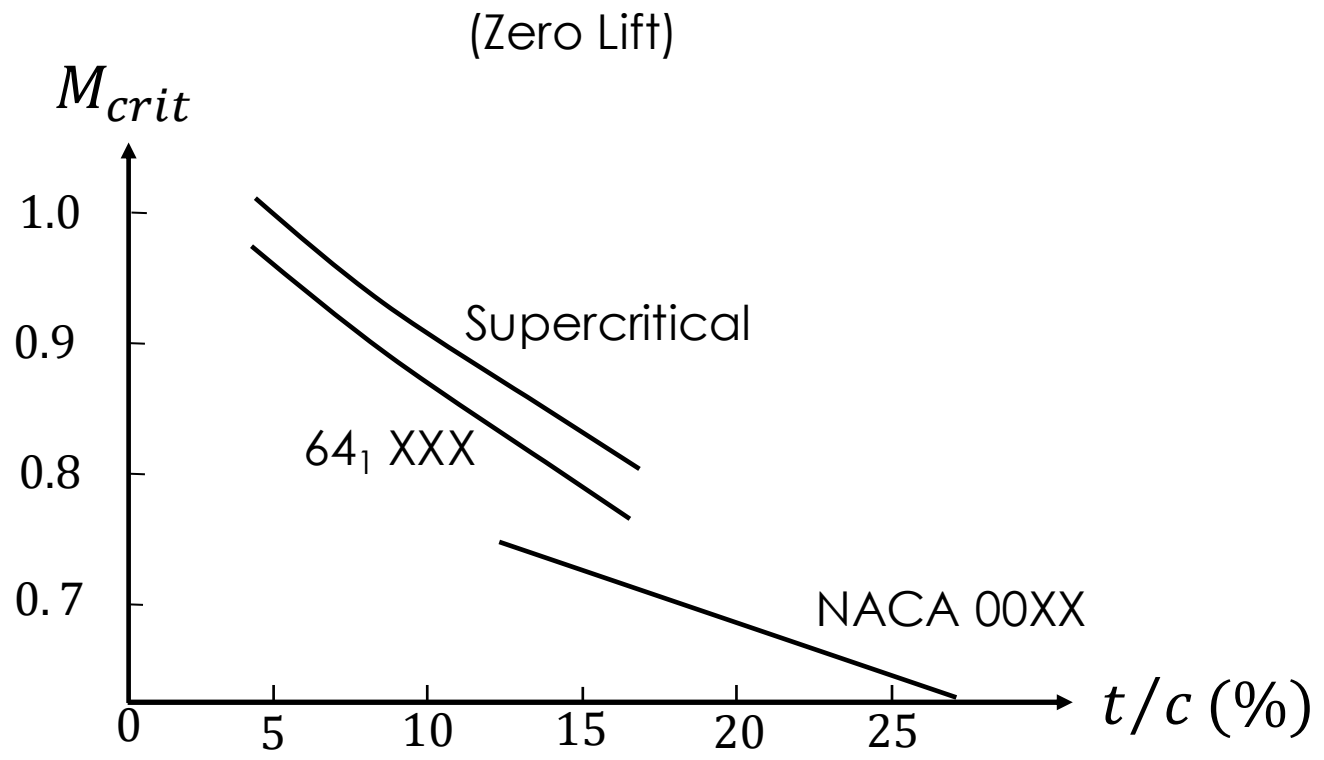
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- As the thickness ratio increases, the critical Mach Number decreases.



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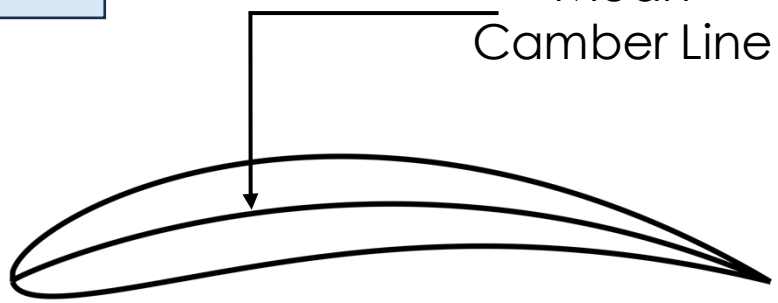
Notes

- Thickness also affects the structural weight of the wing.
- Statistical equations for wing weight:
  - varies approximately inversely with the square root of the thickness ratio.
  - Halving the thickness ratio will increase wing weight by about 41%.
  - The wing is typically about 15% of the total empty weight, so halving the thickness ratio would increase empty weight by about 6%. When applied to the sizing equation, this can have a major impact.

Maximum Lift Characteristics

Camber

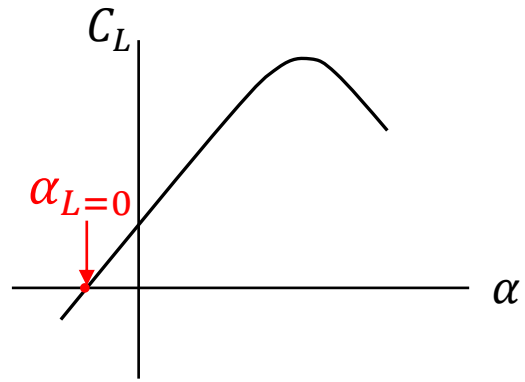
Reminder



Mean Camber Line: is a line joining the mid points between upper and lower surfaces.

What does camber determines?

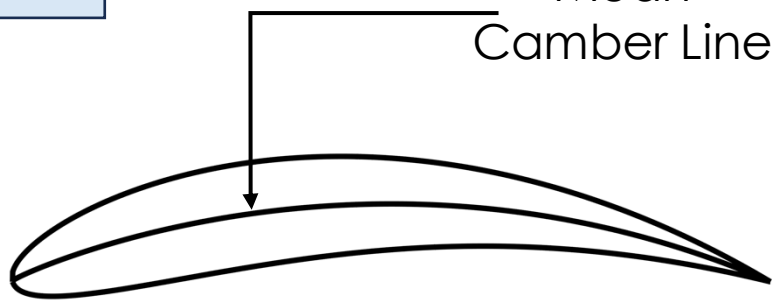
- Camber - this curvature determines how much lift is generated at zero angle of attack.



Maximum Lift Characteristics

Camber

Reminder

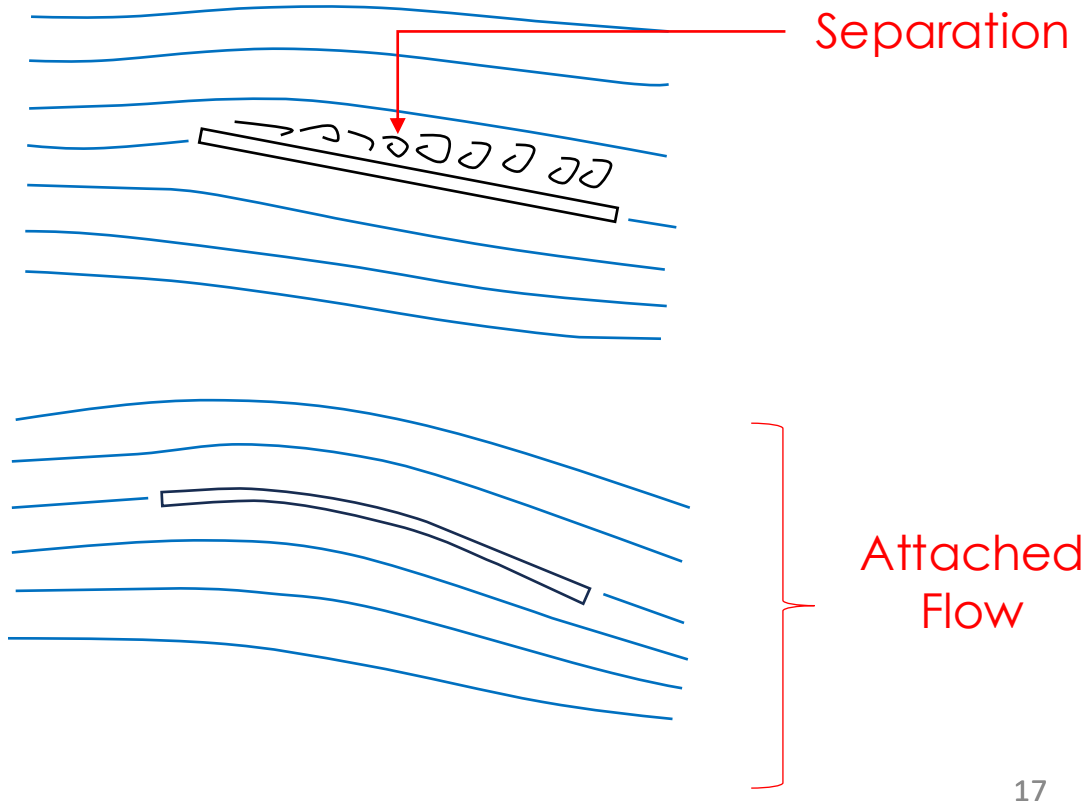


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  - In general, the addition of camber is always beneficial to $C_{l_{max}}$ and the benefit grows with increasing camber.

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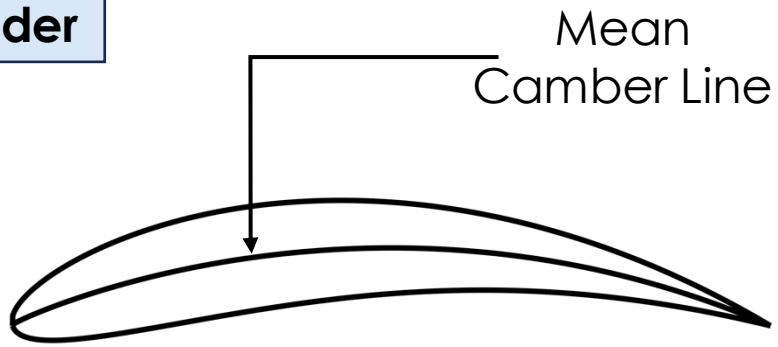
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Maximum Lift Characteristics

Camber

Reminder



Mean Camber Line: is a line joining the mid points between upper and lower surfaces.

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  - In general, the addition of camber is always beneficial to $C_{l_{max}}$ and the benefit grows with increasing camber.

- The camber line shape
  - Position of maximum camber produces higher values of $C_{l_{max}}$.

What does camber determines?

- Camber - this curvature determines how much lift is generated at zero angle of attack.

Note:

- Camber is more effective on thin sections than on thick sections.

Maximum Lift Characteristics

Leading Edge Radius and Leading Edge Shape

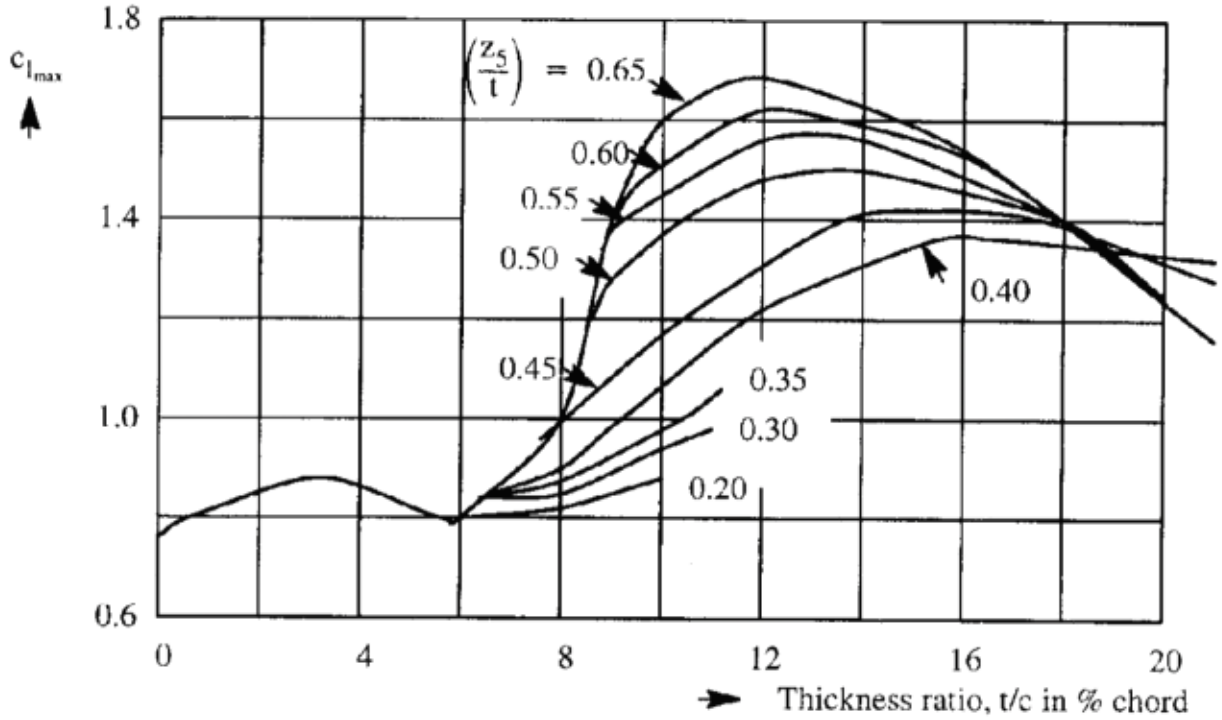
Reminder

Leading edge radius



- The effect of leading edge radius and leading edge shape is more or less reflected in a geometric parameter, called: $\frac{z_5}{t}$.
  - z_5 is the local thickness of the aerofoil at 5% chord and t is the maximum thickness of the aerofoil.
  - A large value of $\frac{z_5}{t}$ indicates a large leading edge radius.

NACA Symmetrical aerofoils at Reynolds Number of $6\ 10^6$.

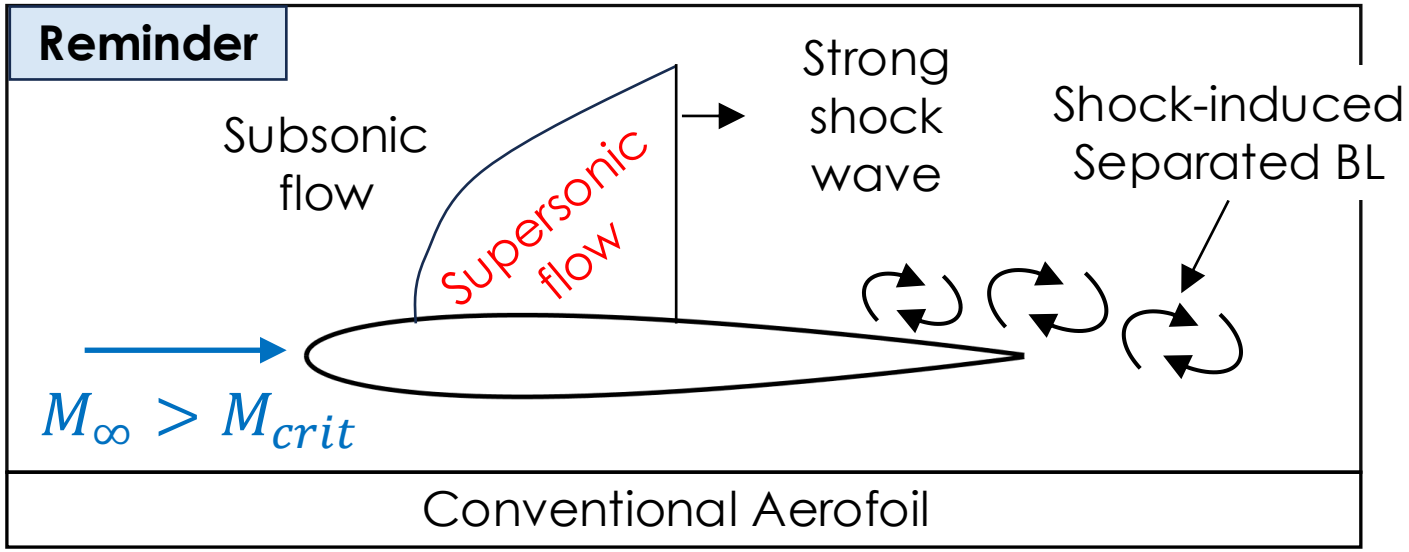


Notes:

- Large leading-edge radii are beneficial in producing large values of $C_{l_{max}}$ at low speeds.
- If supersonic, sharp (or nearly-sharp) leading edge to prevent bow-shock.

Other Aerofoil Considerations

Supercritical Aerofoils

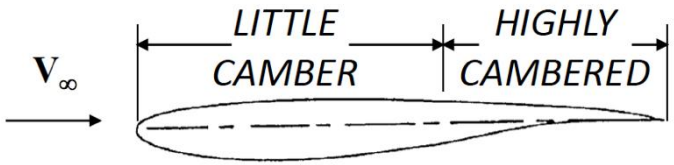


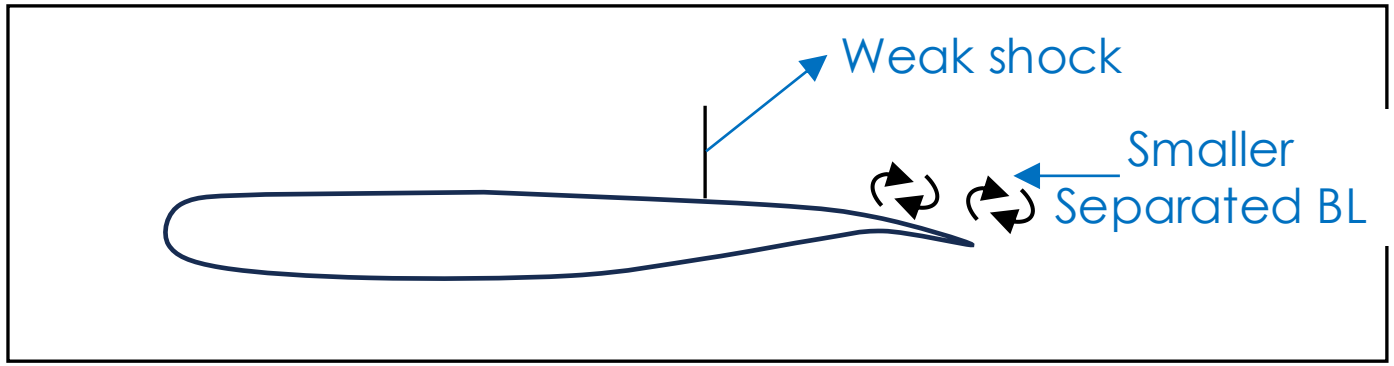
Notes on Conventional Aerofoil:

- If the airplane is flying at just under the speed of sound, the faster air traveling over the upper surface will reach supersonic speeds causing a shock to exist on the upper surface.
- The speed at which supersonic flow first appears on the aerofoil is called the "critical Mach number" $\mathrm{M_{crit}}$.
- The upper-surface shock creates a large increase in drag, along with a reduction in lift and a change in the pitching moment. The drag increase comes from the tendency of the rapid pressure rise across the shock to thicken to even separate the boundary layer.

Other Aerofoil Considerations

Supercritical Aerofoils



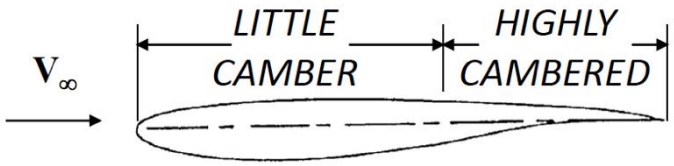


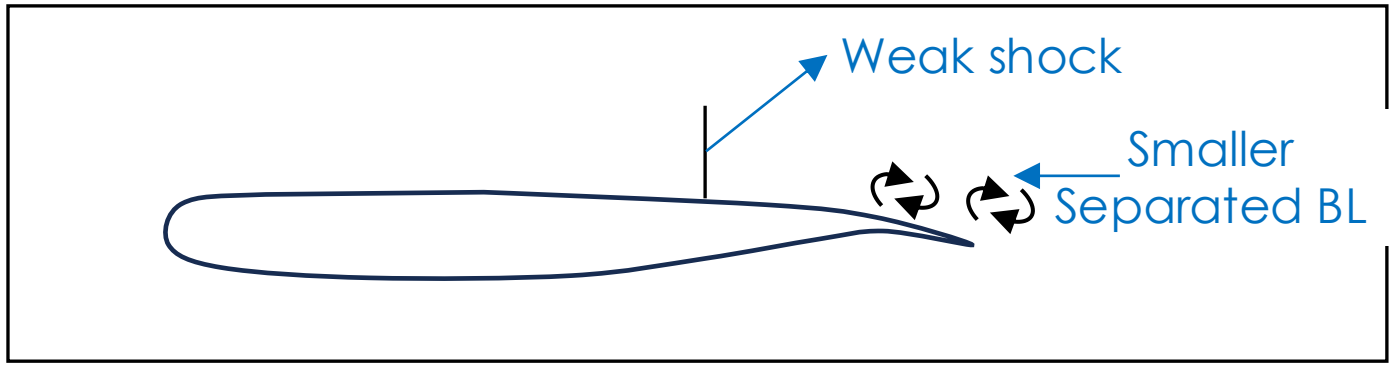
Notes:

- The design of airfield in which the upper-surface shock is minimized or even eliminated by spreading the lift in the chord wise direction, thus reducing the upper surface velocity for a required total lift. This increases the Critical Mach Number.

Other Aerofoil Considerations

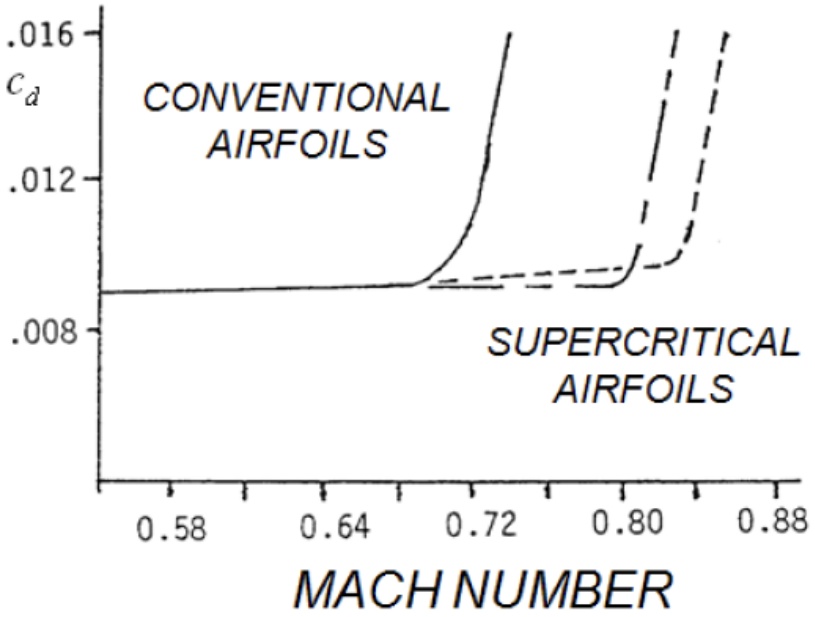
Supercritical Aerofoils





Notes:

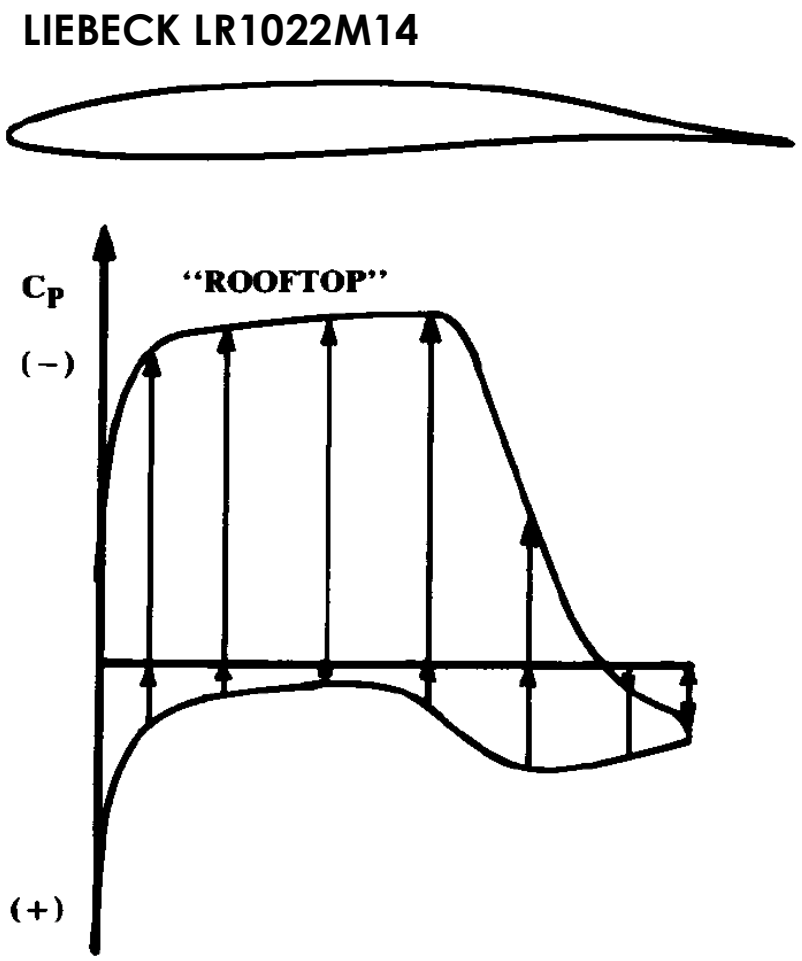
- The design of airfield in which the upper-surface shock is minimized or even eliminated by spreading the lift in the chord wise direction, thus reducing the upper surface velocity for a required total lift. This increases the Critical Mach Number.
- Supercritical aerofoils designed to delay and reduce transonic drag rise due to both strong normal shock and shock-induced boundary layer separation.



Relative to conventional, supercritical aerofoil has reduced amount of camber, increased leading edge radius, small surface curvature on suction side, and a concavity in rear part of pressure side.

Other Aerofoil Considerations

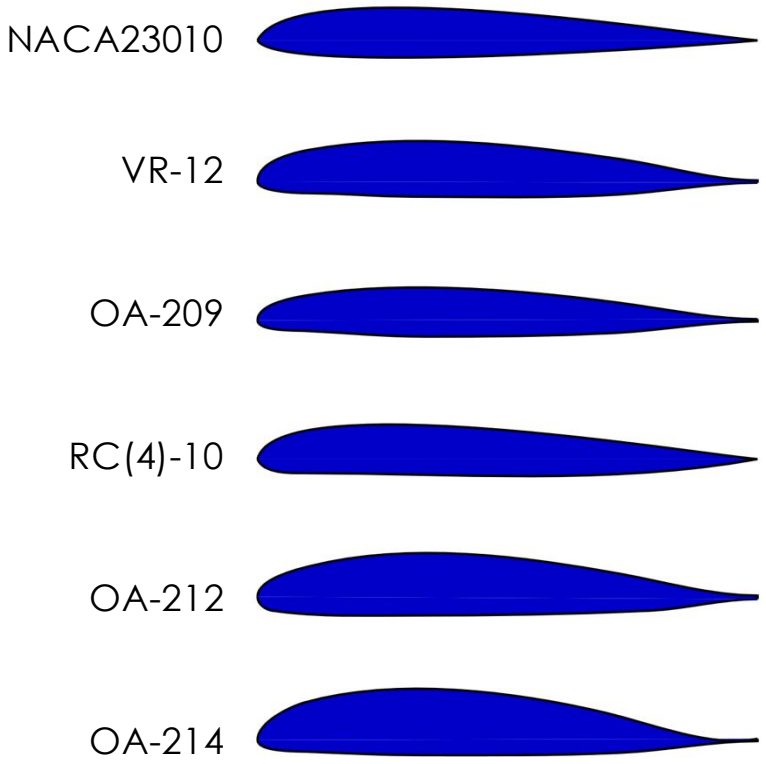
Laminar Aerofoils



- Maintain a laminar flow over the greatest part of the aerofoil. Laminar flow can be maintained by providing negative pressure gradient , i.e. by having pressure drop continuously from leading-edge to trailing edge. This tends to suck the flow towards rearward, promoting laminar flow.
- Laminar flow produces lower skin friction and drag on the airfoil.
- The downside is that such airfoils typically produce lower maximum lift coefficients. Such airfoils are also very sensitive to the surface finish, which must be glassy-smooth and free of contamination to realize the low "laminar" drag values.
- A good laminar-flow aerofoil combined with smooth fabrication methods can produce a wing with laminar flow over about 50–70% of the wing.

Other Aerofoil Considerations

Rotorcraft Aerofoils



- These airfoil sections address the unique aerodynamic challenges of rotorcraft:
  - Operation at high angles of attack near stall
  - Transonic flow
  - The need for both high lift and low drag, and low pitching moments.
- Improve hovering capabilities by generating high lift coefficients.
- Additionally, they are optimized to maximize their lift-to-drag ratios, improving the overall efficiency and speed of the rotorcraft, particularly during forward flight.
- The need for low-pitching moments on blades means that some airfoils are distinctly reflexed.
- Some airfoils in the series are also optimized for transonic flow conditions, which can occur at rotor blade tips.

Summary

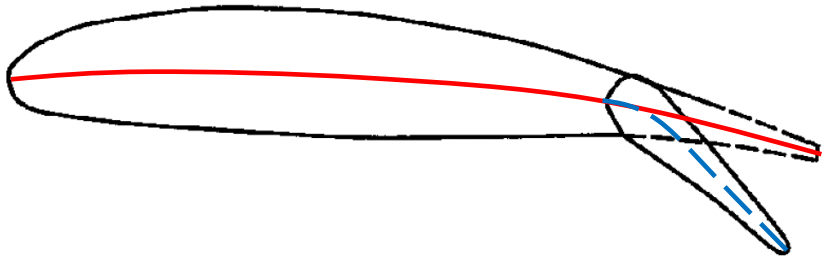
- Various airfoil sections have been geometrically tailored to achieve the best aerodynamic performance in specific flight conditions, such as cruise flight at the highest possible airspeed and Mach number.
- Thicker, more cambered airfoils with rounded noses are better suited to slower flight speeds and low Mach numbers.
- Thin airfoils with sharp leading edges are ideal for high speeds and supersonic Mach numbers.
- Supercritical airfoil sections have been developed for transonic flight conditions, where many airliners operate. These sections reduce wave drag and prevent boundary-layer separation behind the shock wave, enabling airliners to cruise more efficiently at speeds closer to the speed of sound.
- Helicopter airfoils, or rotor blades, face unique aerodynamic challenges because of the widely varying flow conditions they encounter, necessitating designs that optimize performance in both hover and forward flight.

Effect of High Lift Devices

- **Trailing Edge Flaps**
- **Leading Edge Devices**
- **Boundary Layer Control**

Effect of High Lift Devices

- **Trailing Edge Flaps**



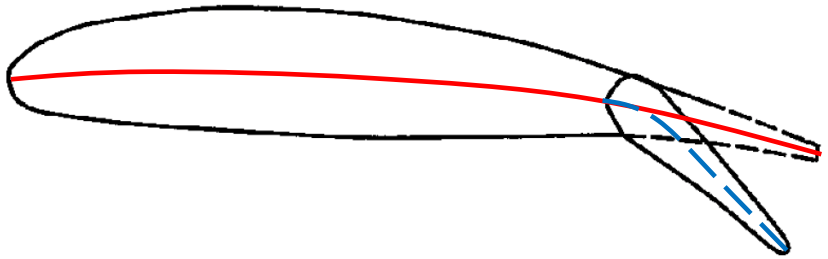
Plain Flap

Plain Flap:

- Formed by hinging the rear-most part of a wing section about a point within the contour.
- By definition, a downward deflection is positive, an upward deflection is negative.
- The main effect of flap deflection is an increase in the effective camber of the airfoil.

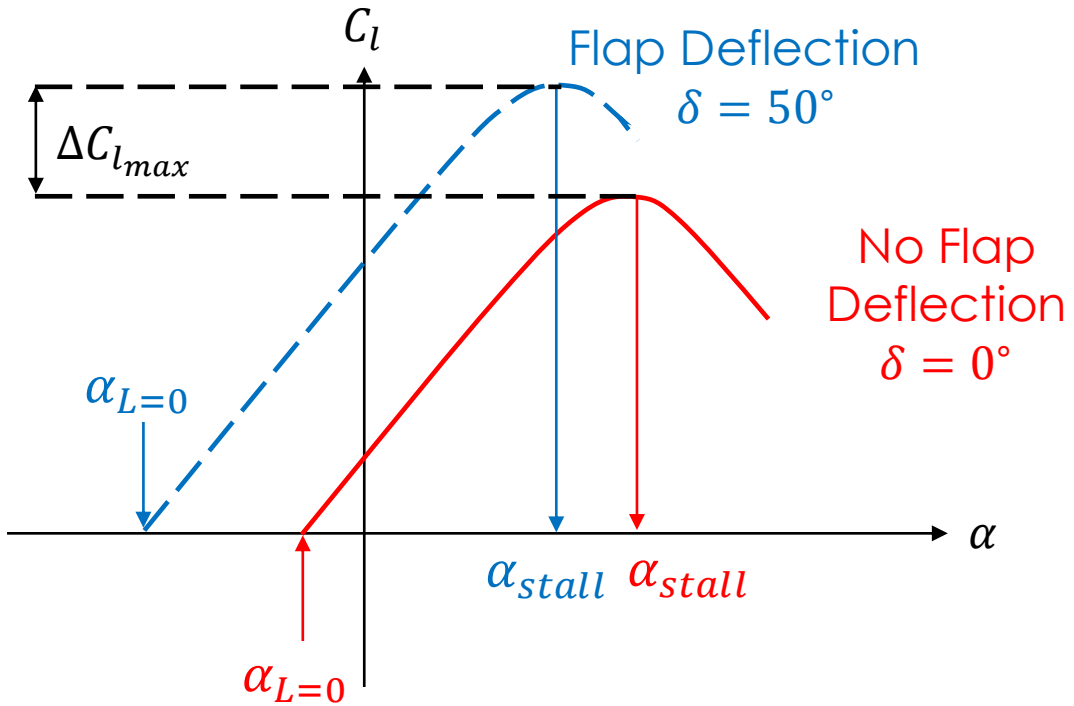
Effect of High Lift Devices

- **Trailing Edge Flaps**



Plain Flap

Split Flap

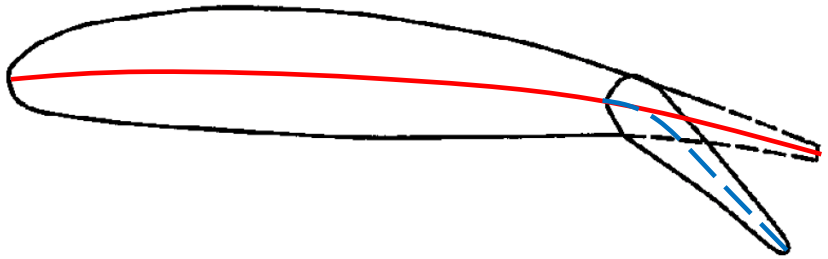


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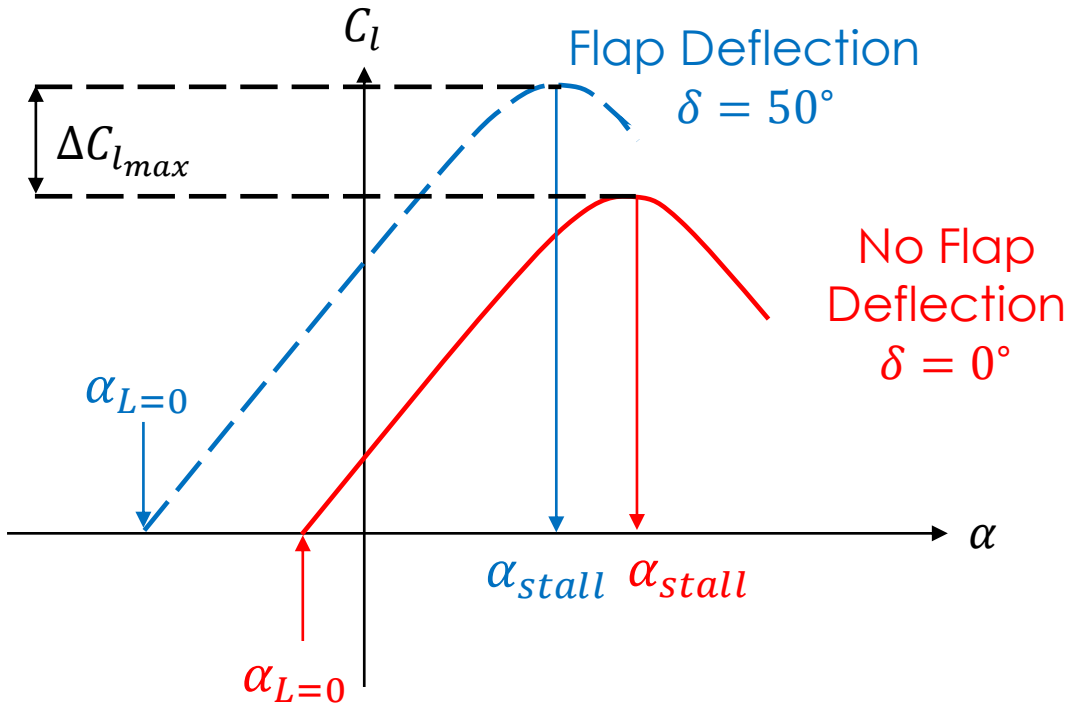
Effect of High Lift Devices

- **Trailing Edge Flaps**



Plain Flap

Split Flap

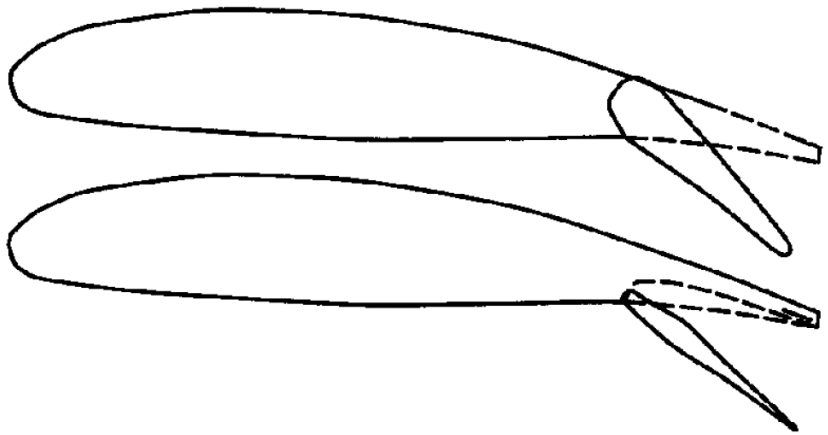


Plain Flap:

- Formed by hinging the rear-most part of a wing section about a point within the contour.
- By definition, a downward deflection is positive, an upward deflection is negative.
- The main effect of flap deflection is an increase in the effective camber of the airfoil.
- It is seen that the flap makes the angle of attack for zero lift much more negative without significantly affecting the lift-curve-slope.
- The stall angle of attack is reduced.
- The drag coefficient is increased and the lift-to-drag ratio is reduced.
- For most commonly used wing sections, the flap deflection angle can be increased to about 15 degrees without flow separation.
- Many single engine, light airplanes use plain flaps.

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Plain Flap

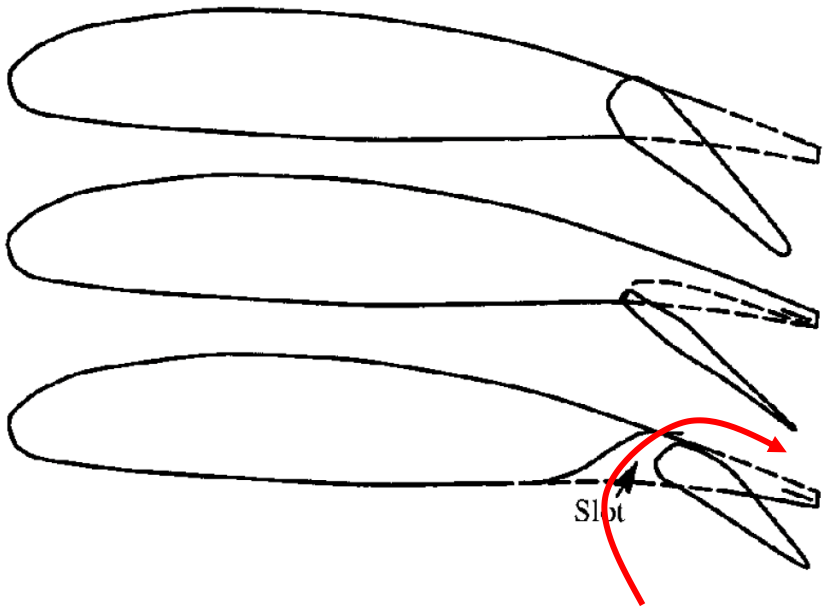
Split Flap

Split Flap:

- Split flaps derive their effectiveness from the large increase in camber produced. Because the upper surface is less cambered than the lower surface with the flap deflected, the separation effects on the upper surface will be less marked than those associated with a plain flap.
- Therefore, the flap performance at high angle of attack is improved.
- The increment in $c_{l_{max}}$ is larger than with a plain flap.
- The drag associated with a split flap is very much higher than that associated with a plain flap because of the large wake.

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- **Trailing Edge Flaps**



Plain Flap

Split Flap

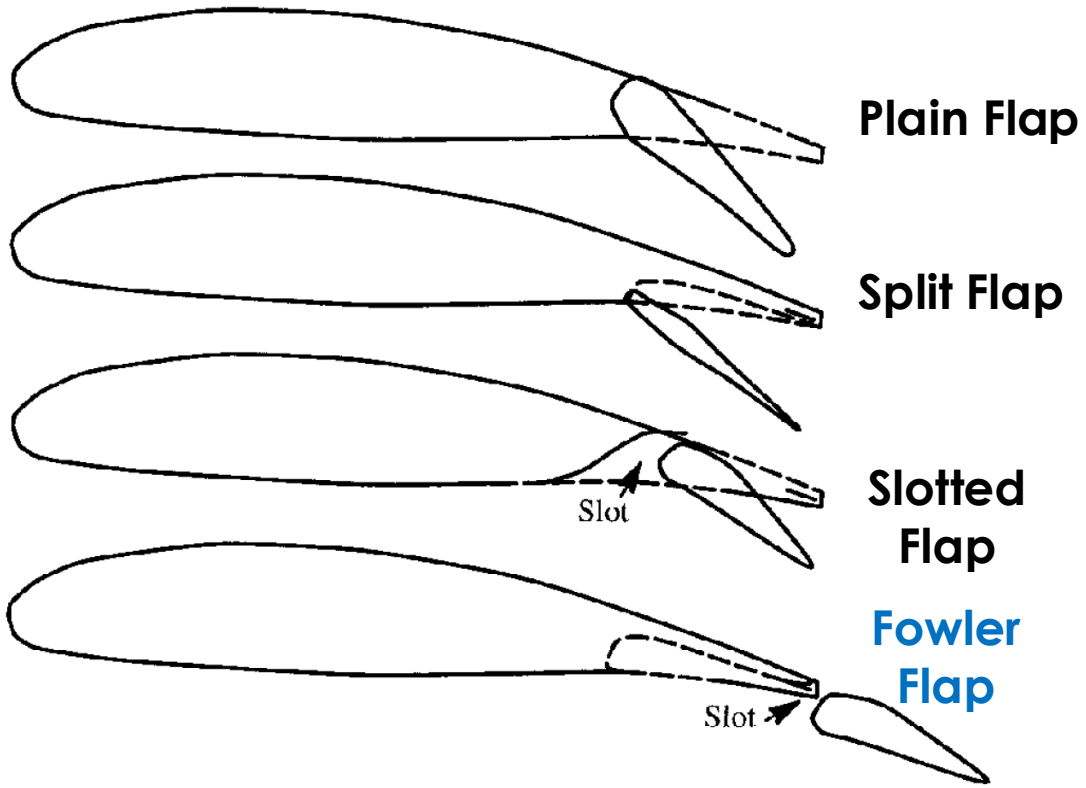
Slotted Flap

Slotted Flap:

- The slot ducts high energy air from the lower surface to the upper surface and direct this air in such a manner as to delay flow separation over the flap by providing boundary layer control.
- The high lift is again derived from the increase in camber which is characteristic for a slotted flap arrangement. The increment in $c_{l_{\max}}$ is much higher than that associated with either a split or a plain flap while the drag is much lower, due to the boundary layer control effect.
- However, the pitching moment associated with slotted flaps tends to be high and negative.

Effect of High Lift Devices

- **Trailing Edge Flaps**

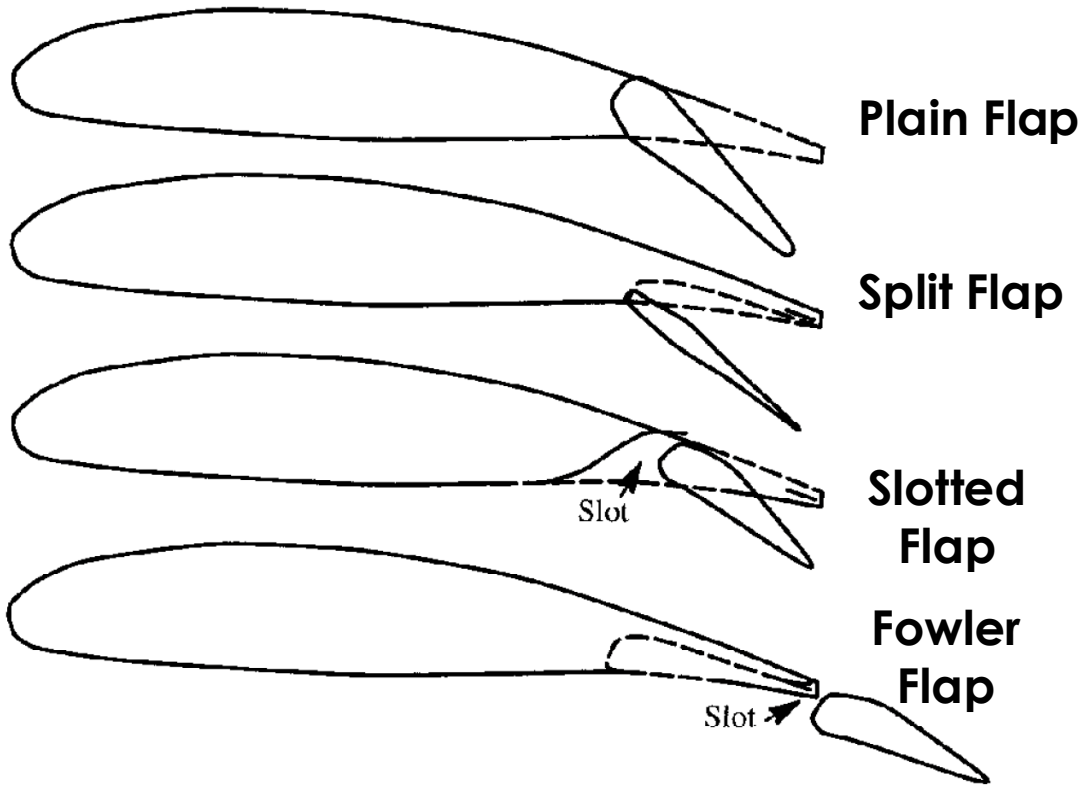


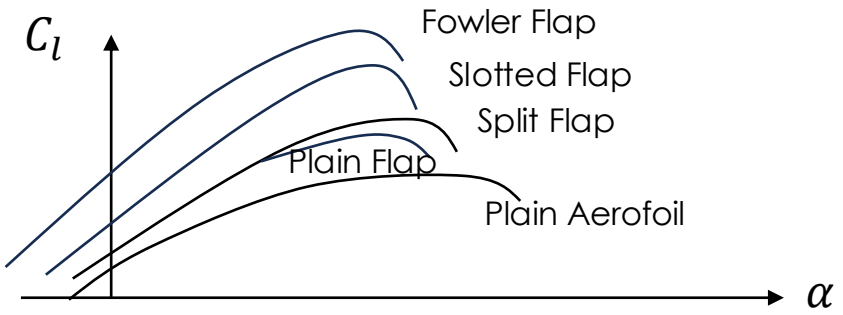
Fowler Flap:

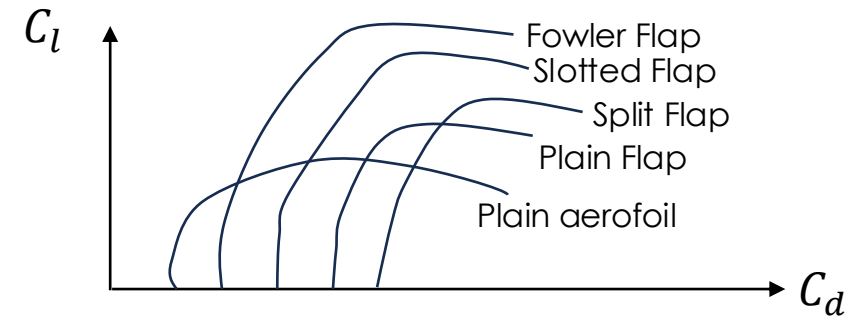
- The Fowler flap employs the same principle as the slotted flap, but it also moves backward while deflecting downward. The backward motion increases the effective wing area.

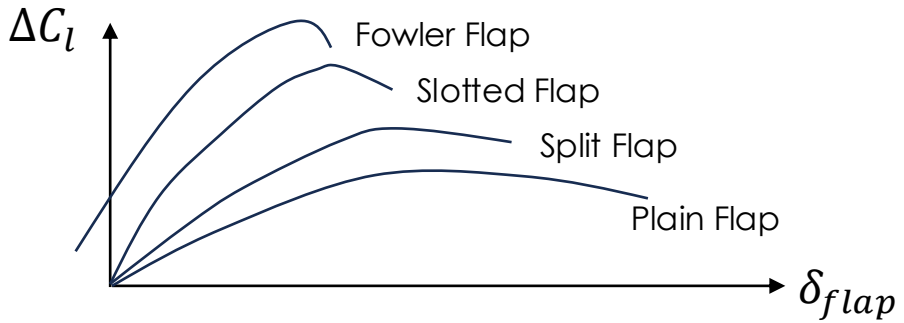
Effect of High Lift Devices

- **Trailing Edge Flaps**





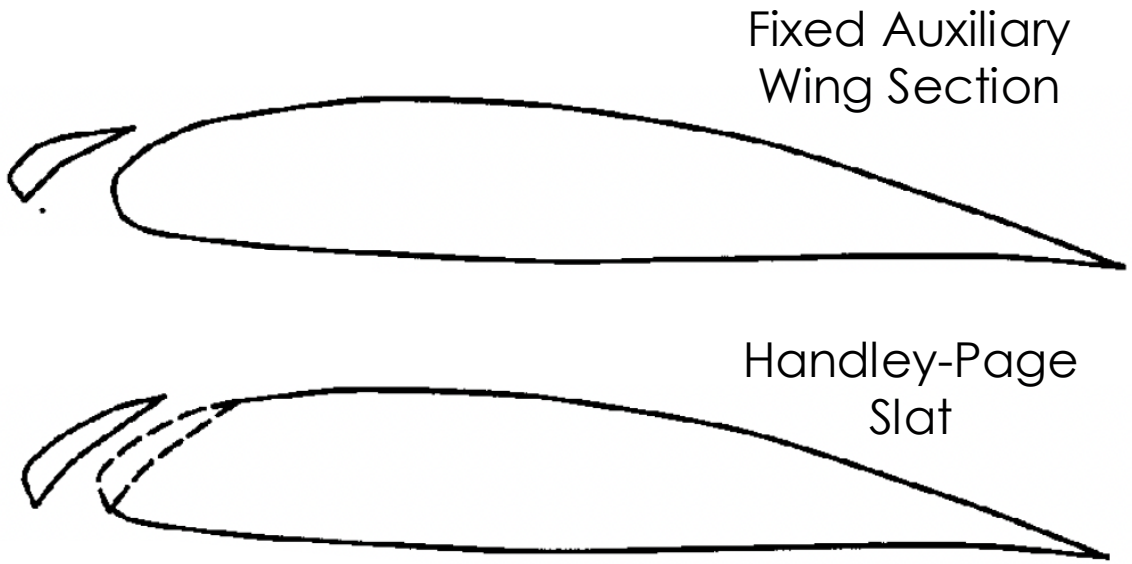




Effect of High Lift Devices

Leading Edge Devices

Slats



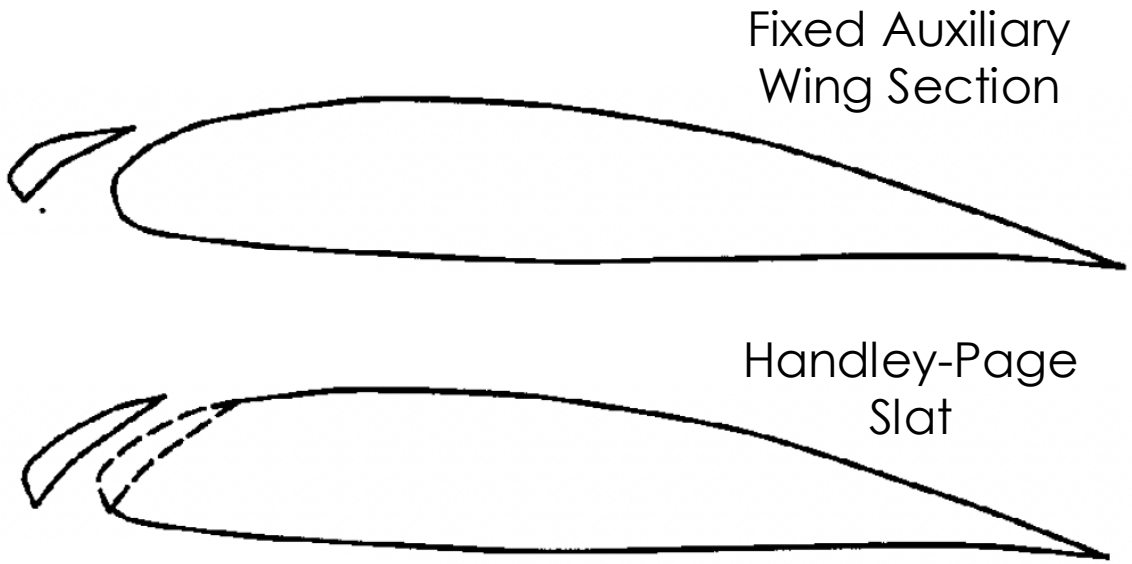
Slats:

- Leading edge slats consists of aerofoils mounted ahead of leading edge of the wing such as to assist in turning the air around the leading edge at high angles of attack and therefore delay leading-edge stalling.
- These slats may either be fixed or retractable.
- It may increase the maximum lift coefficient by as much as 0.5.

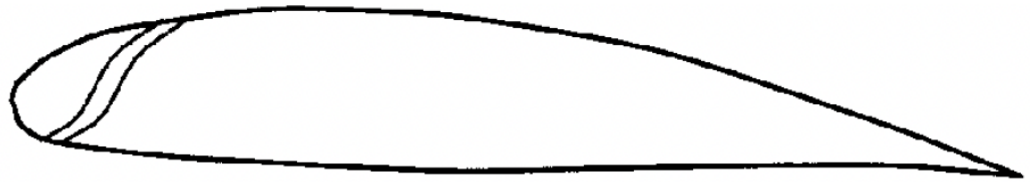
Effect of High Lift Devices

Leading Edge Devices

Slats



Slots



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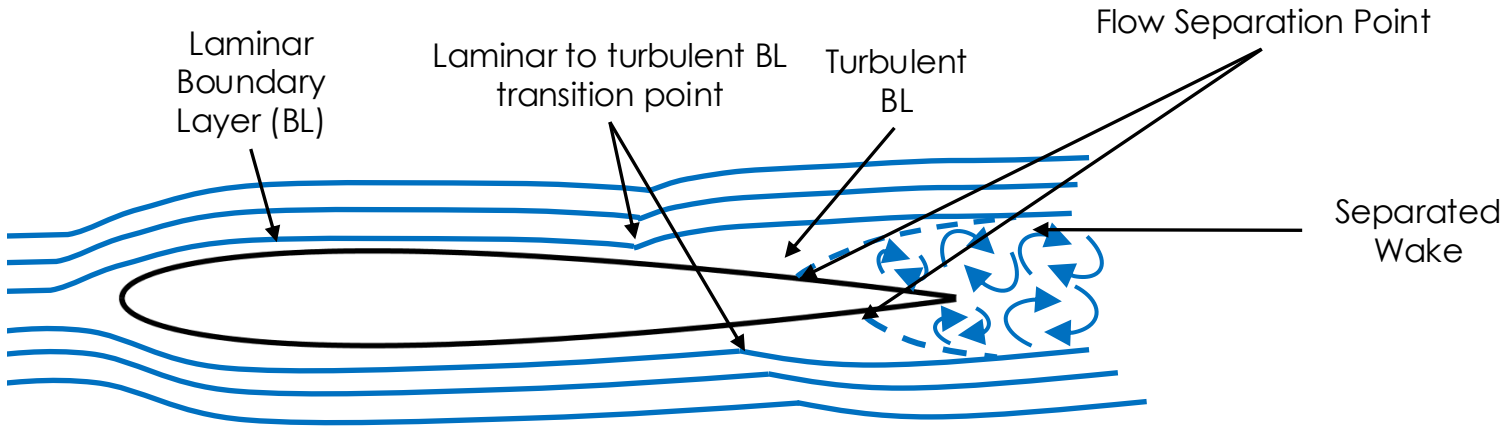
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Slots:

- When the slot is located near the leading edge, this leading edge device differs only in detail from the leading edge slat.
- Additonal slots maybe introduced may be introduced at various chordwise stations.
- The effectiveness of slot derives from BL control effect.
- At low angle of attack, the minimum section profile drag may be greatly increased due to slots.

Boundary Layer Control

- Higher maximum lift coefficients can also be achieved by boundary layer control (BLC).
- BLC may involve injecting high speed air (or gas) called "blowing," or removing the low energy boundary layer by "suction," or both.
- Suction effect can be provided by using a suitable suction pump to suck the shear layer through a porous airfoil skin. This type of boundary layer control can not only increase the lift, but also delay transition to reduce the skin friction drag.



Effect of High Lift Devices

Summary

- High-lift devices are small aerofoil-like elements that are fitted at the trailing edge of the wing as a flap and/or at the LE as a slat. In typical cruise conditions, the flaps and slats are retracted within the contour of the aerofoil.

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- Considerable lift enhancement can be obtained by incorporating high-lift devices at the expense of additional drag and weight.